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Similarity Thresholds in Retrieval-Augmented Generation

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Abstract— This paper evaluates the performance of open-source Large Language Models (LLMs) Mistral:7b, Llama2:7b, and Orca2:7b within the context of Retrieval-Augmented Generation (RAG). The purpose of the study is to identify the similarity score thresholds that yield the best performance across Natural Language Processing metrics. Despite the advancements in LLMs, their ability to access up-to-date, domain-specific information remains limited. RAG addresses this by integrating external database querying to produce contextually comprehensive and precise responses. This study investigates the similarity score thresholds for the LLMs using a custom-developed dataset from the domain of agriculture. Tests were conducted using the PaSSER web-based application, which was enhanced to support detailed configuration of retrieval processes. The PaSSER App is a complementary project to the Smart Crop Production Data Exchange (SCPDx) platform. The results showed that Mistral performs best at lower similarity threshold (0.55), Orca achieves the highest performance at a moderate threshold (0.65), and Llama at a lower threshold (0.55), but has strong performance at higher thresholds as well (0.70 and 0.85). The study discusses several factors that could cause the differences in the similarity score thresholds of the assessed LLMs. Future development will focus on leveraging other pre-trained open source LLMs (over 40b), exploring approaches to fine-tuning, and further integration on PaSSER App with the existing infrastructure of the SCPDx platform.

Keywords- Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Similarity Score Thresholds, Mistral:7b, Llama2:7b, Orca2:7b, Composite Performance Score.

I. INTRODUCTION

The development of Retrieval-Augmented Generation (RAG) systems has marked an important milestone in the field of natural language processing (NLP). Even though Large Language Models (LLMs) have advanced capabilities in categorizing, producing, and understanding text, they often lack specific domain understanding and the ability to access up-to-date information. This limitation presents a significant challenge, which RAG addresses, by integrating prompt engineering and external database querying to produce responses that are both contextually comprehensive and precise, as outlined in [1].

Despite these advancements, the implementation of RAG systems is not without challenges. In study [2] are identified seven critical failure points that must be addressed to realize the full potential of RAG systems: missing content, missed top ranked documents, not in context - consolidation strategy, not extracted, wrong format, incorrect specificity, and incomplete.

The purpose of this paper is to identify the similarity score thresholds that yield the best performance of three LLMs – Mistral:7b, Llama2:7b, and Orca2:7b within Retrieval-Augmented Generation (RAG) across nine NLP metrics – METEOR, Rouge-1.f, Rouge-l.f, BLEU, Laplace Perplexity, Lidstone Perplexity, Cosine similarity, Pearson correlation, F1 score.

Similarity score thresholds are values used in information retrieval systems to determine the relevance of documents or objects by comparing similarity scores, which measure the likeness between items. It sets the minimum score required for an item to be considered relevant. Similarity thresholds are often discussed in the context of optimizing retrieval performance. For example, thresholds are adjusted to balance precision and recall, ensuring that retrieved items are both relevant and diverse [3].

The study employs the PaSSER web-based application [4] to automate the testing and analysis process. It is motivated to explore how to enhance the overall effectiveness and reliability of RAG using open-source LLMs.

The rest of the paper is organized as follows. The next section reviews related work. Section III describes the PaSSER web application used for testing and analysis. Section IV outlines the data preparation and testing procedure. Section V introduces the Composite Performance Score (CPS) as a means to identify the similarity score thresholds for the LLMs. Section VI presents the evaluation and results. Section VII discusses the various reasons for the variation in similarity score thresholds among the LLMs considered. Section VIII concludes the research and outlines future directions.

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II. Related Work

Studies such as [5], [6], [1], and [2] provide comprehensive overviews of RAG systems, highlighting their effectiveness in enhancing the accuracy and contextual relevance of generated responses by integrating external knowledge sources. These studies also discuss the technical challenges, including the need for efficient data retrieval mechanisms and the integration of diverse data types.

In the field of comparative analyses, the researches [7] and [8] delve into the comparative performance of various RAG methods, such as full fine-tuning and parameter-efficient fine-tuning, versus retrieval-augmented approaches. These analyses help in understanding how different methodologies stack up in terms of accuracy, computational efficiency, and applicability to various NLP tasks. The paper [9] also contributes to this category by benchmarking the performance of various large language models in RAG tasks, providing insights into their strengths and weaknesses across different scenarios.

In [10] and [11] are explored implementation of advanced RAG techniques, including the use of knowledge graphs and dense passage retrieval, which further enhance the retrieval accuracy and the quality of the generated content. These studies provide insights into state-of-the-art implementation strategies that address common limitations like data sparsity and retrieval speed.

Improvements in query optimization and context management, is highlighted in [12], contributing to the efficiency and effectiveness of RAG systems. These advancements ensure that the retrieved information is highly relevant and contextually appropriate for the generated responses.

A detailed survey [13] on the progression of RAG paradigms highlights state-of-the-art technologies embedded in RAG frameworks. It offers an overview of the advancements and trends in RAG systems.

These studies underline the progress and ongoing challenges in the development of RAG systems. They highlight the importance of efficient data retrieval, diverse data integration, and continuous assessment and enhancement of RAG methodologies to improve the accuracy and contextual relevance of generated responses.

III. Web Application

The PaSSER App is a complementary project to the Smart crop production data exchange (SCPDx) platform, described in detail in [14] and [15]. The platform aims to support the integration of exchanging information and data acquired or generated as a result of the use of different technologies in smart agriculture. The PaSSER Web App is described in [4].

It is an open-source application, available for download at: https://github.com/scpdxtest/PaSSER.

For performing the tests in this paper, the configuration capabilities of PaSSER have been enhanced to provide users with more control over the LLMs and retrieval from vector database processes.The Fig. 1 provides a structured layout of the features and navigation options available in Web App. The new features are included in the `Configuration` panel. To the main options for choosing the path for the Ollama, selecting the model, the ChromaDB, and setting the temperature is added switch for Retriever options.

Query processing in the PaSSER App follows two primary workflows. The first is standard Q&A chat. It directly processes user queries through the LLM to generate responses. The second is RAG Q&A chat, where a user queries with context retrieved from the vectorstore before processing by the LLM.

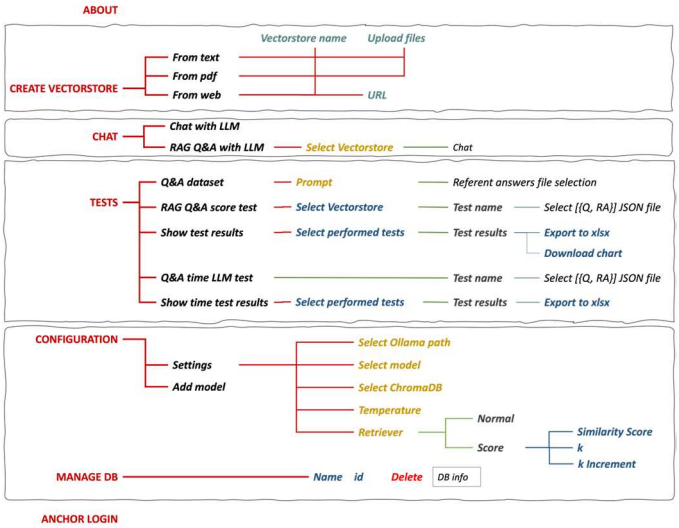

Figure 1. A site map of the PASSER Web App.

The RAG Q&A chat workflow now includes a **retriever switch** with two modes: Normal and Score. The Normal mode utilizes the standard LangChain retriever, while the Score mode allows users to configure parameters such as similarity score, k, and k Increment to fine-tune the retrieval process for more precise results.

Normal Mode: Utilizes the standard LangChain retriever, invoking `vectorStore.asRetriever()`.

Score Mode: Enables users to set values for parameters similarity score, k, and k Increment [16].

A *minSimilarityScore* sets the minimum similarity score threshold for documents to be considered relevant. It ranges from 0 to 1, where higher values ensure only highly relevant documents are retrieved.

A *maxK* determines the maximum number of documents to retrieve. It limits the retrieval to a manageable number, with our setup ensuring that up to 100 documents can be considered, aligning with the size of our dataset.

A *kIncrement* defines the increment size for document retrieval in each iteration. Setting this at 2 allows for fine-grained control, ensuring the process is thorough without being overwhelming.

IV. Data Preparation

The selected LLMs are Mistral:7b [17], Llama2:7b [18], and Orca2:7b [19]. These models were chosen for their balance

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between performance and affordability, making them suitable for medium computational capacity environments. Mistral:7b features attention mechanisms like grouped-query attention and sliding window attention, optimizing processing speed and handling longer text inputs efficiently. Llama2:7b is designed for a broad range of text-based tasks, employing an auto-regressive language model framework optimized through supervised fine-tuning and reinforcement learning. Orca2:7b is a fine-tuned version of Llama2, enhancing reasoning abilities with high-quality synthetic data. These models have the same volume parameters and are suitable to be installed and used on hardware configurations with medium computational capacity, while giving good enough results.

The knowledge base used was retrieved from Climate-smart agriculture Sourcebook (https://shorturl.at/tdSza). The knowledge base is pre-processed manually and vectorized using the transformers of Misrtal:7b, Llama2:7b, and Orca2:7b LLMs under the following parameters: chunk size—500, overlapping—100, temperature—0.2.

In this study, a subset of 101 questions/answers pairs is used from the previously created dataset of 446 questions [4]

The currently used dataset is available at: GitHub link. A total of 3,030 tests were performed (3 models, 10 similarity score values, and 101 questions).Testing is conducted on Mac Mini M1 with 16 GB RAM and a 10-core GPU.

Evaluations were performed at different similarity scores (0.5 to 0.95 in increments of 0.05).

The metrics used: METEOR, Rouge-1.f, Rouge-l.f, BLEU, Laplace Perplexity, Lidstone Perplexity, Cosine similarity, Pearson correlation, F1 score.

METEOR (Metric for Evaluation of Translation with Explicit Ordering) [20] evaluates alignment considering precision, recall, synonymy, stemming, and alignment. ROUGE (Recall-Oriented Understudy for Gisting Evaluation) [21] measures overlap of n-grams, word sequences, and pairs. BLEU (Bilingual Evaluation Understudy) [22] assesses accuracy and fluency compared to reference texts. Perplexity (PPL) [23] indicates model uncertainty in predicting the next word. Cosine Similarity [24] measures similarity between query and retrieved documents. Pearson Correlation assesses linear correlation between predicted and actual values. F1 Score [25] combines precision and recall into a single metric.

These metrics are selected to provide an evaluation of the models, covering various aspects of performance such as text quality, retrieval accuracy, coherence, and predictive accuracy. Each metric contributes to a detailed understanding of how well the models perform in different scenarios, ensuring a thorough and robust assessment.

V. Composite Performance Score

In order identify the similarity score thresholds that yield the best performance of LLMs, a Composite Performance Score (CPS) is compiled. This reduces the complexity of comparing models by combining multiple metrics into a single score. The CPS helps with similarity score thresholds selection without having to go into the details of individual metrics.

Different metrics can be weighted according to their importance for a particular application or domain.

The weights assined to each metric based on their importance are set as follows:

- METEOR: 15%
- ROUGE-1, ROUGE-l: 15% (7.5%, each)
- BLEU: 15%
- Perplexity (Laplace and Lidstone): 15% (7.5% each, with lower values indicating better performance)
- Cosine Similarity: 10%
- Pearson Correlation: 10%
- F1 Score: 20%

High weights (15-20%) are assigned to metrics that provide an overall assessment of text quality that predict performance and accuracy (METEOR, BLEU, F1 Score, Perplexity).

Moderate weight (10%) is assigned to metrics that measure relevance and retrieval correlation (Cosine Similarity, Pearson Correlation).

Smaller weights (7.5% each) are assigned to ROUGE metrics to account for different levels of similarity (overlap) of the text.

The weights can vary depending on the specific application and priorities of the study. For example, if the main focus is on text generation quality, metrics like METEOR and BLEU might receive higher weights. However, it is obvious to suggest that a balanced approach that considers both retrieval accuracy and text generation quality is preferable to provide a more comprehensive evaluation.

These weights were assigned to balance the different aspects of model performance and to capture and evaluate the key aspects as much as possible. Priority is given to the most comprehensive and impactful metrics.

For each of the similarity thresholds, a CPS is calculated for each of the models using the weighted average of the normalised metrics.

The CPS calculation Python code is as follows:

```
import argparse
import pandas as pd

# Example code to calculate composite score
metrics = ['METEOR', 'Rouge-1.f', 'Rouge-2.f', 'Rouge-l.f', 'BLEU', 'Laplace Perplexity', 'Lidstone Perplexity', 'Cosine similarity', 'Pearson correlation', 'F1 score']
weights = {'METEOR': 0.15, 'Rouge-1.f': 0.075, 'Rouge-2.f': 0.0, 'Rouge-l.f': 0.075, 'BLEU': 0.15, 'Laplace Perplexity': 0.075, 'Lidstone Perplexity': 0.075, 'Cosine similarity': 0.10, 'Pearson correlation': 0.10, 'F1 score': 0.20}

def normalize(series):
   return (series - series.min()) / (series.max() - series.min())

def composite_score(df, weights):
   normalized_df = df[metrics].apply(normalize)
```


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```
        print("IN COMPOSITE SCORE\n", normalized_df)
        score = sum(normalized_df[metric] * weight for metric, weight in weights.items())
        print("SCORE in DEF\n", score)
        return score

    parser = argparse.ArgumentParser(description='Calculate normalized scores.')
    parser.add_argument('file', help='The Excel file to read.')
    args = parser.parse_args()

    df = pd.read_excel(args.file, engine='openpyxl')

    df = pd.read_excel(args.file)
    df_selected = df.iloc[0:100, 5:21]

    #print("DF", df)
    df_to_normalize = df.iloc[2:102, 6:21]
    normalized_df = df_to_normalize.apply(normalize)
    df.iloc[2:102, 6:21] = normalized_df

    #print(df)

    # Calculate the composite score
    composite_score = composite_score(df_selected, weights)
    print ("COMPOSITE SCORE\n", composite_score, composite_score.mean())
```

The code is executed from the console with the command `python3 normalize.py <filename>.xlsx`, for each of the files containing the raw test data.

Raw test data is available at: GitHub link.

VI. Evaluation and Results

Here is a summarized table with the CPS estimated for Mistral, Orca, and Llama LLMs across different similarity thresholds:

TABLE I. COMPOSITE PERPFRMANCE SCORES (SPC) SUMMARY

Similarity score	Mistral composite score	Orca composite score	Llama composite score
0.5	0.463150	0.483298	0.458887
0.55	0.487886	0.488232	0.491925
0.6	0.463460	0.505304	0.411094
0.65	0.460408	0.526650	0.483548
0.7	0.468361	0.500202	0.490521
0.75	0.424785	0.472629	0.484744
0.8	0.463281	0.518104	0.475556
0.85	0.482175	0.526259	0.484952
0.9	0.468986	0.521275	0.489074
0.95	0.481841	0.491266	0.416642

Key Observations could be summarized as follows. Mistral achieves its highest CPS of approximately 0.487886 at a similarity score of 0.55. It performance is relatively stable across different thresholds, with a noticeable dip at 0.75 (0.424785) and slight improvement towards higher thresholds. The performance fluctuates moderately, indicating that Mistral can handle a range of similarity scores but performs best at lower to moderate thresholds.

Orca achieves its highest CPS of approximately 0.526650 at a similarity score of 0.65. It shows a clear upward trend in performance as the similarity score increases from 0.5 to 0.65. The performance peaks at 0.65 and slightly decreases at higher thresholds but remains relatively high. Orca consistently performs well across the majority of similarity score thresholds, demonstrating robustness and versatility.

Llama achieves its highest CPS of approximately 0.491925 at a similarity score of 0.55. The model exhibits significant variability in performance across different thresholds, with notable dips at 0.6 and 0.95. Besides the peak at 0.55, Llama also shows relatively high performance at 0.7 and 0.85.

Orca demonstrates the most stable performance across a range of thresholds, making it a reliable choice for various settings. Mistral and Llama show more fluctuations, suggesting they might require more careful tuning to achieve better performance.

The variation in composite scores across different similarity thresholds indicates that each model responds differently to changes in the threshold.

The chart on Fig. 2. provides a visual comparison of the performance of LLMs - Mistral, Orca, and Llama at varying similarity score thresholds.

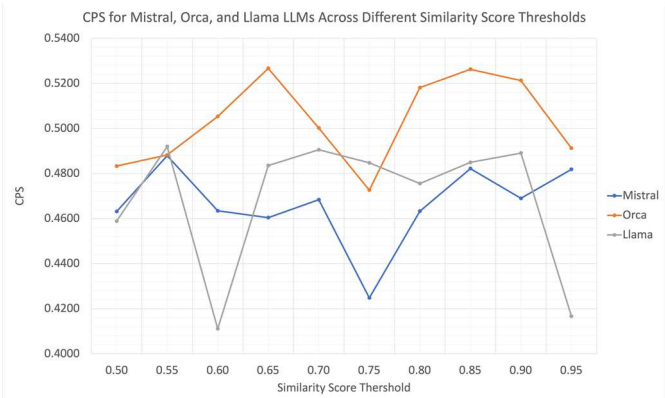

Figure 2. Composite Scores for LLMs across similarity score thresholds.

The chart confurms that different models are sensitive to different similarity score thresholds.

The results may have some implications in terms of model selection. For instance, if the application requires robust and stable performance across a wide range of similarity scores, the Orca LLM would be a suitable choice. Conversely, Mistral and Lama LLMs could be preferred for applications favoring low to moderate similarity scores, with specific tuning around the 0.55 threshold for optimal results.

The results suggest that further studies could be conducted using additional tests with more refined increments of similarity points around the identified peaks to refine the optimal thresholds. The effect of different training datasets and fine-tuning techniques on the variability of Mistral and Llama performance should be investigated.

By interpreting the CPS trends and variations at different similarity score thresholds, this analysis helps to understand how

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to improve the performance of different LLMs in the RAG. This will ensure that the models and settings chosen are consistent with the desired application outcomes.

VII. Discussion

The variation in similarity score thresholds for the Mistral, Orca, and Llama LLMs in RAG can be attributed to several factors, which are well-documented in recent scientific publications [3], [26], [27]. These factors include model architecture, training data, context sensitivity, retrieval strategies.

Table II summarizes the factors that might affect the similarity score thresholds for Mistral, Orca and Llama LLMs.

TABLE II. FACTORS INFLUENCING SIMILARITY SCORE THRESHOLDS FOR LLMS

Factor	Mistral	Orca	Llama
Model Architecture and Training Data	Moderate sensitivity due to diverse training data	Balanced architecture and data handling	Requires highly relevant data due to training specificity
Context Sensitivity	Benefits from moderately relevant context	Handles broader context well	Needs highly relevant context for best performance
Retrieval and Generation Balance	Prefers balance between diversity and relevance	Excels with balanced retrieval and generation	Performs best with highly relevant but less diverse context
Noise Handling and Robustness	Robust to some noise, can handle moderate relevance	Moderate noise tolerance	Sensitive to noise, prefers high relevance for quality
Similarity Calculation Method	Moderate impact due to balanced relevance	Sensitive to balance between relevance and context richness	High impact, requires strict relevance criteria
Generalization Capabilities	Strong generalization, moderate context suffices	Good balance between generalization and specificity	High specificity required
Training and Fine-Tuning Differences	Trained on diverse data, hence moderate threshold	Balanced training and fine-tuning	Highly specific training, hence high threshold needed
Domain and Query Specificity	Performs well across diverse domains	Adaptable to various domains	Requires domain-specific fine-tuning for best results
Context Utilization Efficiency	Utilizes moderately relevant context effectively	Efficient in broader context utilization	Highly efficient with strictly relevant context
Retrieval Strategy	Moderate relevance strategy	Balanced relevance strategy	High relevance strategy

The similarity score threshold affects the balance between *retrieving* highly relevant document chunks form vector database and *generating* high-quality responses. Lower thresholds might retrieve more diverse but less relevant documents, while higher thresholds ensure high relevance but might miss some useful context. Different models might have different strengths in handling this balance, leading to varied thresholds.

Models with better *noise-handling* capabilities might perform well with lower thresholds as they can filter out irrelevant information during generation. Models less robust to noise might need higher thresholds to ensure only highly relevant documents are retrieved, leading to better overall performance.

The way in which *similarity scores* (e.g., cosine similarity in vector space) are calculated and interpreted by different models can significantly influence the similarity score thresholds considered appropriate for retrieving relevant documents. Variations in these calculations and interpretations mean that each model may perform best at different thresholds, emphasizing the need for model-specific tuning to achieve best performance in retrieval-augmented generation tasks.

The observed differences in similarity thresholds for Mistral, Orca and Llama can be attributed to their architectures and training approaches.

Mistral's architecture incorporates grouped query attention and sliding window attention mechanisms. These features allow it to efficiently process longer contexts, potentially enabling it to extract relevant information from even moderately relevant documents. Its training on diverse datasets likely contributes to its ability to generalise well, allowing it to perform optimally at a lower threshold where it can exploit a wider range of contextual information.

As a fine-tuned version of Llama2, Orca's optimal performance at a moderate threshold suggests a balanced approach to exploiting context. Its training on high quality synthetic data for enhanced reasoning skills may allow it to effectively exploit moderately to highly relevant information. The 0.65 threshold is likely to provide an optimal balance between retrieval diversity and relevance for Orca's reasoning.

Llama2's architecture is designed for a wide range of text-based tasks, which may explain its variability in performance at different thresholds. Its strong performance at both lower and higher thresholds suggests a flexible ability to exploit both diverse and highly relevant information. The model's supervised fine-tuning and reinforcement learning during training may contribute to this adaptability, allowing it to perform well in different retrieval scenarios.

These model-specific characteristics highlight the importance of considering architectural differences and training methods when optimising RAG systems. Future research could include a more detailed analysis of how specific architectural features (e.g., attention mechanisms, model size) and training data characteristics influence optimal similarity thresholds in RAG applications.

Mistral might have strong generalization capabilities, making it perform better with a moderate amount of highly relevant context.

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Orca likely benefits from a balance of relevance and context richness, performing best at a higher threshold.

Llama may require highly relevant context to generate the best responses, leading to maximum CPS at the highest threshold.

Different training datasets and fine-tuning techniques can make models more or less sensitive to the amount and relevance of context provided during retrieval.

The *specific domain or type of queries used for testing* might influence the similarity threshold, as different domains might require different levels of context specificity.

VIII. Conclusion and Future Directions

The study provides an evaluation of three open-source LLMs: Mistral:7b, Llama2:7b, and Orca2:7b within the context of RAG. The primary goal was to identify similarity thresholds that maximize performance across multiple NLP.

The research identifies specific similarity score thresholds that yield the best composite performance scores for each model: Mistral performs best at lower similarity threshold (0.55), Orca achieves the highest performance at a moderate threshold (0.65), and Llama at a lower threshold (0.55), but has strong performance at higher thresholds as well (0.70 and 0.85). These findings could help in fine-tuning RAG in order to achieve the best balance between retrieval relevance and text generation quality.

In order to identify the similarity score thresholds that yield the best performance of LLMs, a Composite Performance Score (CPS) is compiled. The CPS is a weighted average of normalized scores of model performance on a set of metrics.

The study employs the open-source web-based application PaSSER to automate the testing and analysis process. The paper details the enhancements made to PaSSER by introduction of a retriever switch with configurable parameters (similarity score, k, and k increment), which allows for fine-tuning the retrieval process.

The paper highlights that the variation in similarity score thresholds between models can be attributed to different factors such as model architecture, training data, context sensitivity, model robustness to noise and generalization capabilities.

Future development will focus on leveraging other pre-trained open source LLMs (over 40b), exploring approaches to fine tuning and further integration on PaSSER App with the existing Antelope blockchain/IPFS infrastructure on the SCPDx platform.

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